Task 3b. Physical Design of Data Warehouse

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Different Design Fundamentals

A. What have we done so far in the Design?

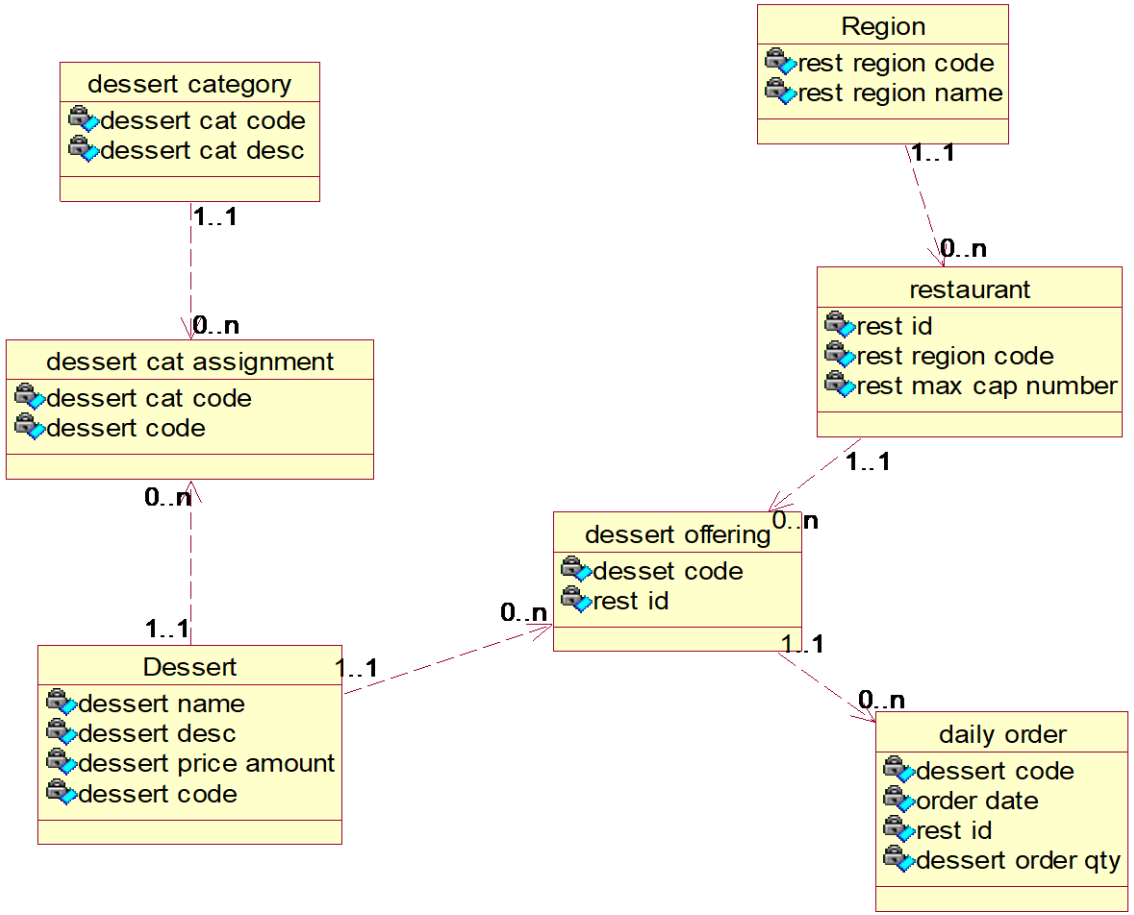
- Conceptual Design (use ERD or Dimensional Modeling) on paper
- Logical Design (Translates to a RDBMS)

- **Physical Model –** adding physical characteristics like indexing style, etc.
- **Implementation model --** use of SQL Server specifics

Physical Design

It is a way for understanding the data requirements within a given context or environment.

LOGICAL DESIGN FUNDAMENTALS



Logical Design (ERD)

It is a way to understand the overall data requirements.

Physical Design Fundamentals

B. What is a Physical Design?

- The physical design transforms the logical design to a particular OS, a DBMS, a hardware and a platform.
- The major objectives of the physical design are:

1. Improve Performance
   - Response in OLTP is < 3 secs, while in OLAP is less stringent
   - Response still should be very quick even if we are processing a very large volume of data

Physical Design Fundamentals

B. What is a Physical Design?

2. Ensure Scalability
   - Usage increases over time
   - Usage increases due to number of users increases and number of concurrent users increase
3. Manage Storage
   - Proper management of stored data will boost performance

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Physical Design Fundamentals

B. What is a Physical Design?

4. Provide Ease of Administration
   - Methods for proper arrangement of table rows in storage so that frequent reorganization is avoided.
   - Back up and recovery of database tables
5. Design for Flexibility
   - Keep the design open so that the changes to the physical model is achieved.

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Physical Design Fundamentals

C. Steps to create a Physical Design

1. Develop standards
   - Standards ensure consistency across various areas
   - It is very important in the data warehouse world as users can be from other disciplines
   - Naming of Database Objects -- a clear method must be present for composing names of the data warehouse objects.
   - Naming of the files and tables in the staging area
   - Standards for physical files -- source code files, database files and application documents

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Physical Design Fundamentals

C. Steps to create a Physical Design

2. Create Aggregate Plan

- Worry about the summary tables
- If you have data stored in the lowest level of granularity, accessing summery data can involve lot of calculations
- How many summary tables need to be created?
- Need a comprehensive plan for aggregation

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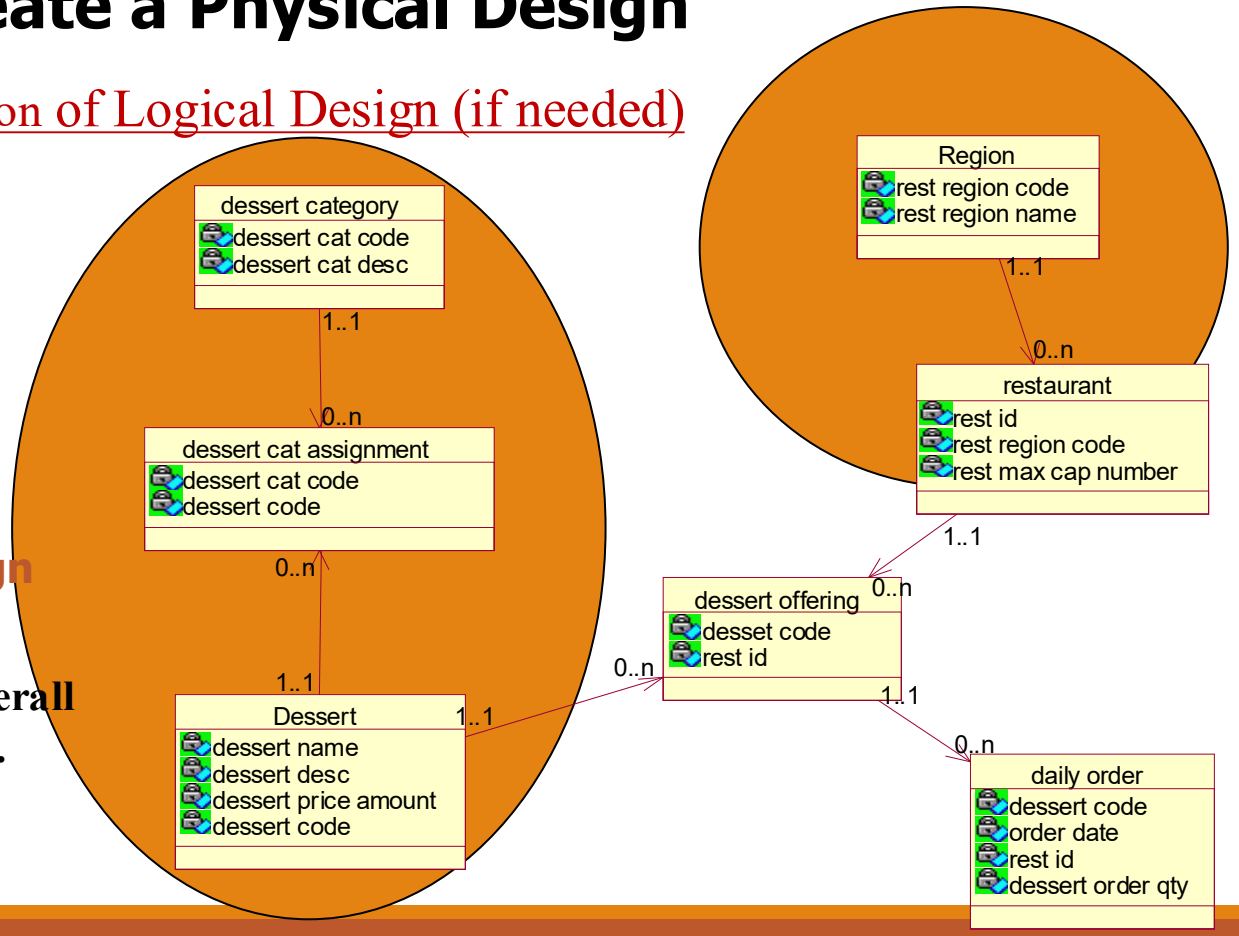
Physical Design Fundamentals

C. Steps to create a Physical Design

3. Denormalization of Logical Design (if needed)



Logical Design

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Physical Design Fundamentals

C. Steps to create a Physical Design

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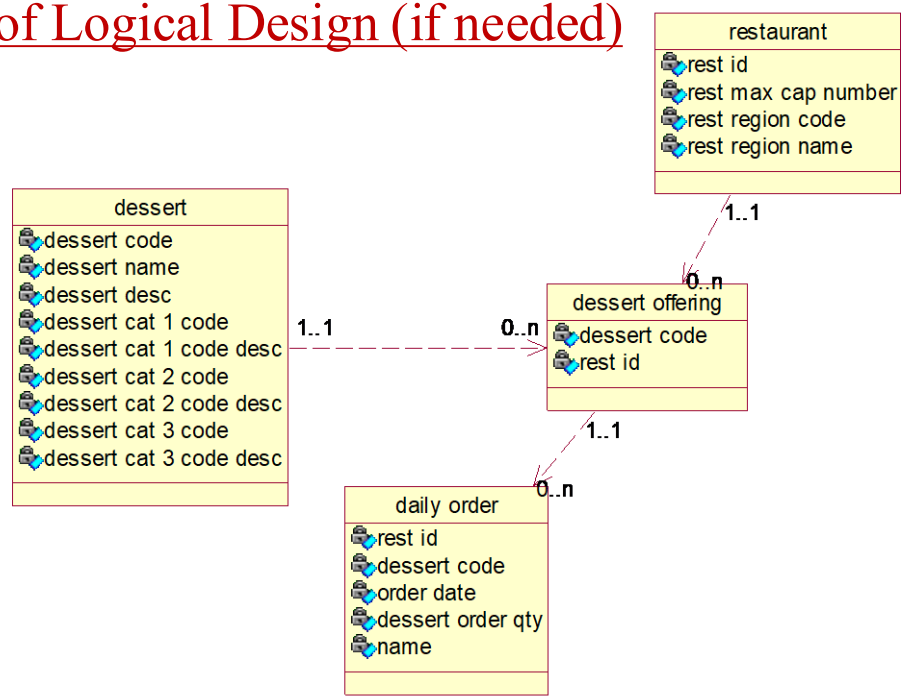

Physical Design Fundamentals

C. Steps to create a Physical Design

4. Determine the Data Partitioning Scheme

- In dimensional world, most dimension tables are small.
- But some DIMENSION TABLES (like Product table) and the FACT TABLES ARE VERY LARGE.
- Large tables are not easy to manage
- In the load process, the entire table needs to be closed to the users
- Backup and recovery are problems for such large tables
- Think about "partitioning" the tables

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Physical Design Fundamentals

C. Steps to create a Physical Design

4. Determine the Data Partitioning Scheme

- The partitioning scheme must include
- Horizontal or vertical?
- Number of partitions for each table
- The criteria for dividing each table
- Description how to make queries aware of partitions

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Physical Design Fundamentals

C. Steps to create a Physical Design

4. Determine the Data Partitioning Scheme

Vertical partitioning: to store Dessert price amount in its own table as it might change much more frequently

Code	Name	Desc	Price	Cat 1 code	Cat 1 desc	Cat 2 code	Cat 2 desc	Cat 3 code	Cat 3 desc
1	Banana split	Several bananas, etc	$4.95	1	Frozen	2	Indulgent		
2	Brownie sundae	Large pieces of brownie, etc.	$3.95	5	Cake	1	Frozen	2	Indulgent
3	Yogurt parfait	Low fat yogurt, etc	$2.95	3	Healthy choice	1	frozen		

Horizontal partitioning if too many rows

Physical Design Fundamentals

C. Steps to create a Physical Design

5. Prepare an Indexing Strategy

- Unlike OLTP system, data warehouse is query-centric
- Create Indexing plan to support query performance
- Indexing is very specific to the DBMS (so please check it)

Indexing the Data Warehouse

(i). Why indexing is important in DW?

- Indexing is important in any DB application
- Indexing is very important in Data Warehouse as it is very "query-centric"
- Types of indexes supported:
  - Index sequential
  - B-Tree indexing
  - Bit-mapped indexing

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Indexing the Data Warehouse

(ii). Indexing Issues in Data Warehouse

- Indexes and Loading
  - When you have many indexes, loading of data in the warehouse slows down
  - The problem is more acute at the time of initial load
  - one solution - kill the indexes before the load and after the load is complete recreate the indexes

Indexing the Data Warehouse

(ii). Indexing Issues in Data Warehouse

- Index-Only Records
  - Remember that any DBMS will read index record first
  - Add columns in the index record to provide for other information so that unnecessary records are not read.
- Selecting Columns for Indexing
  - Examine the queries and see which columns are being used frequently

Indexing the Data Warehouse

(ii). Indexing Issues in Data Warehouse

- Indexing for Large Tables
  - When the table is too large, many indexes for this table can create problems
  - In such cases, consider splitting the table and then create indexes

Indexing the Data Warehouse

(iii). Indexing Types -- B-Tree Index

- B-Tree is a hierarchical levels of "index records"
- The lowest level index records point the rows in the data tables
- If a column has many unique vales, then the selectivity of the column is said to high.
    - In a "sales territory" dimension table, the "city" column has high selectivity and should be used for Indexing.
    - For some columns, the selectivity may not be there. You can make the selectivity of columns higher by concatenating the columns --- but this increases the size of the "index" file

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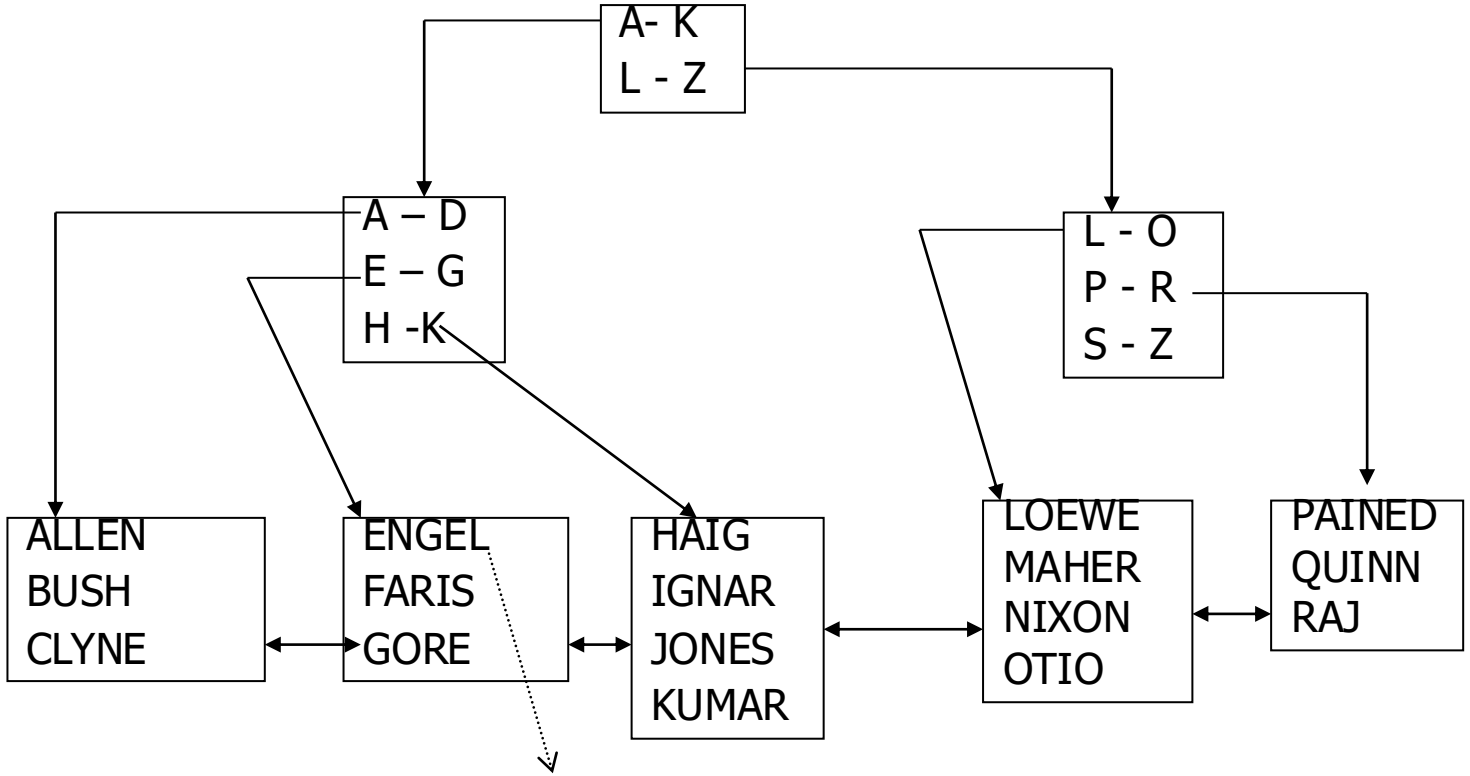
Indexing the Data Warehouse

(iii). Indexing Types -- B-Tree Index



Pointers to data rows

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Indexing the Data Warehouse

(iii). Indexing Type – Bitmapped Index

- Bitmapped is ideally suited for **low-selectivity** data.
- A bitmap is an ordered series of bits, one for each distinct value of the indexed column.
- Example:
  - Product Color – white 100
    almond 010
    black 001
  - Product – dishwasher 001
    washer 100
    dryer 010
  - Region -- east 1000, west 0100
    north 0010, south 0001

bit mapped indexes

1. When a column is said to be **highly selective**, this means that a very low percentage of the total rows possible is returned which is another way of saying that the column contains a lot of unique values. Columns with high selectivity are often the best candidates for indexes.
2. Columns with **low selectivity**, very few unique values, are often not good candidates for indexes as often the index will be ignored in favor of a table scan or clustered index scan as the

Indexing the Data Warehouse

(iii). Indexing Type – Bitmapped Index

		Extract of Sales Data			
Rowid	Date	Product	Color	Region	Sale($)
1	15-Nov-00	Dishwasher	White	East	300
2	15-Nov-00	Dryer	Almond	West	450
3	16-Nov-00	Dishwasher	Almond	West	350
4	16-Nov-00	Washer	Black	North	460
5	17-Nov-00	Washer	White	South	700
6	17-Nov-00	Dryer	White	East	400
7	17-Nov-00	Washer	Almond	South	475

Pick columns that have low selectivity for indexing

(a) Product

(b) Color

(c) Region

Bitmapped Index for Product	
Washer, Dryer, Dishw asher	
1	001
2	010
3	001
4	100
5	100
6	010
7	100

Bitmapped Index for Color	
White, Almond, Black	
1	100
2	010
3	010
4	001
5	100
6	100
7	010

Bitmapped Index for Region	
East, West, North, South	
1	1000
2	0100
3	0100
4	0010
5	0001
6	1000
7	0001

select the rows fromSales
where product="washer'
color ="almond"
division= east or south

Product –100
Color -- 010
Region – 1000 or 0001

Indexing the Data Warehouse

(iv). Indexing Type – Clustered Index

- Does not use a separate data record and segment record
- All index data and real data are in one place
- Improve performance as the disk look up is reduced – one read gets data and index

Indexing the Data Warehouse

(v). Indexing Fact Table

- the primary key of the fact table is the primary keys of all dimension tables
- Create a B-Tree Index on the “full” primary key
- Carefully design the order of the full concatenated key for indexing
- Bitmapped indexes do not apply to fact table as there is no low-selectivity column in the fact table
- etc.

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Indexing the Data Warehouse

(vi). Indexing the Dimension Table

- Columns of the dimensional table are used for the predicates of queries
- Create a unique B-Tree index on single-column primary key
- Examine the columns that are commonly used to constrain queries. These are actually ready for bitmapped indexing.
- Look for columns that are frequently accessed together in large dimension tables – think about multicolumn indexes

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Physical Design Fundamentals

C. Steps to create a Physical Design

6. Assign Storage Structures

- Storage plan must include "temporary data extract files", the staging area, and any storage needed for front-end applications
- Need to be concerned with the physical files also

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Physical Storage

A. Storage Area Data Structures

- Data Staging Area
  - Data Extract Files
  - RDBMS files (with transformed data)
  - RDBS Index files
- Data Warehouse Tables (DB files and Index Files)
- OLAP System files (proprietary matrix format for storing multidimensional cubes)

Physical Storage is important for project

Physical Storage

B. Estimate the Storage Sizes

- For each database table, determine
  - initial estimate of number of rows
  - average length of rows
  - anticipated monthly increase
- For all tables, determine
  - the total number of indexes
  - space needed for indexes
- Estimate
  - temporary file sizes in the data staging area
  - temporary workspace for sorting and merging

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Physical Design Fundamentals

C. Steps to create a Physical Design

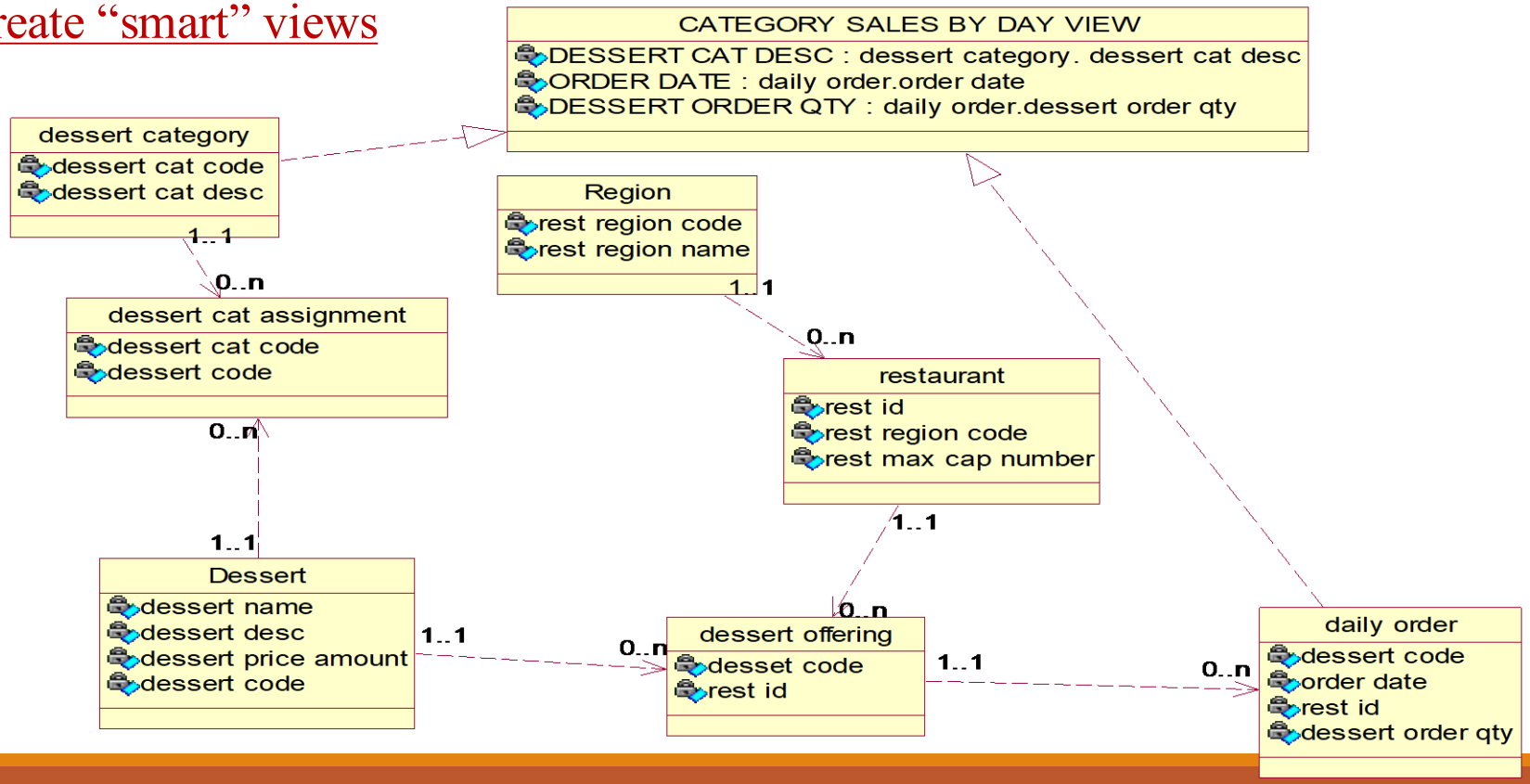
7. Create "smart" views

   - Views can offer the same benefits as denormalization without the drawbacks associated with data redundancy

Physical Design Fundamentals

C. Steps to create a Physical Design

7. Create "smart" views



Physical Design Fundamentals

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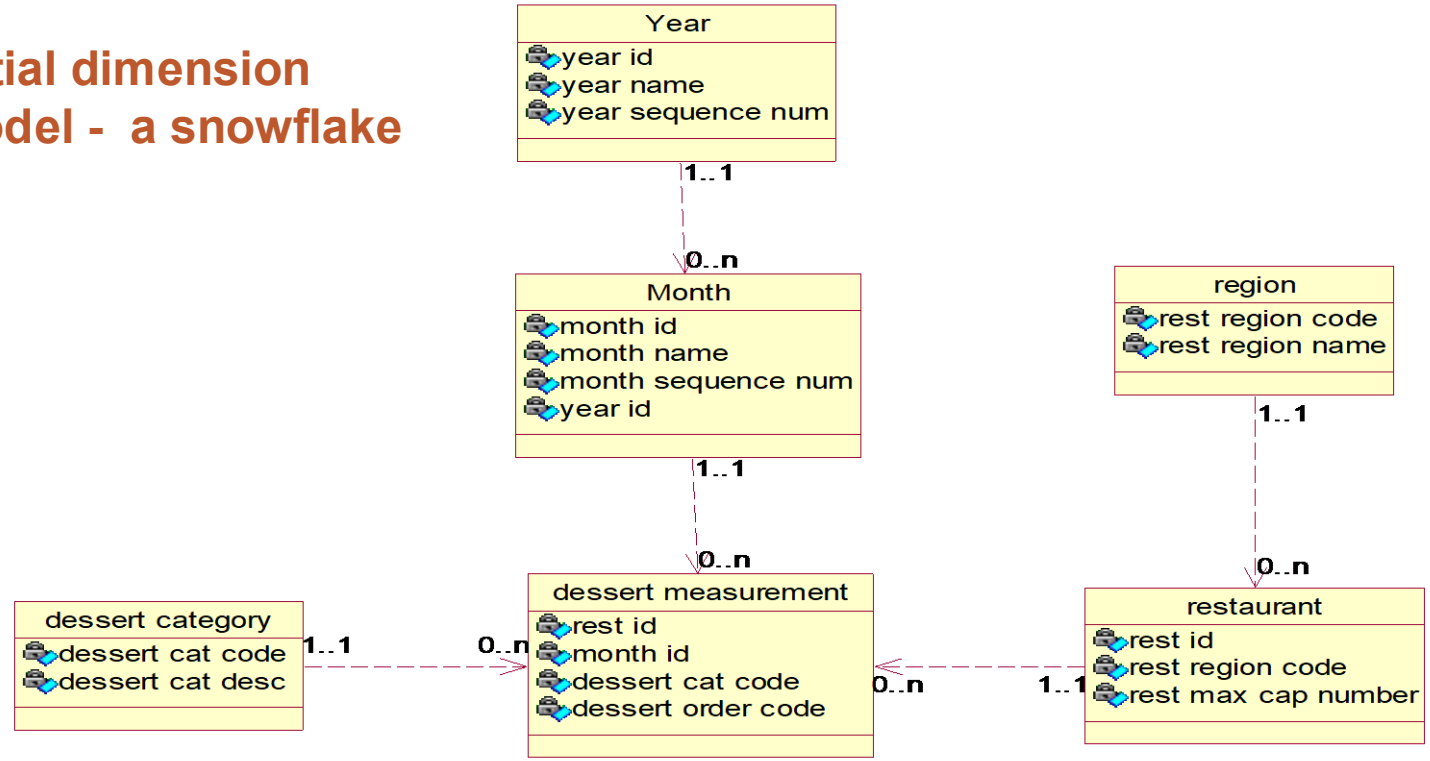
8. Convert Snowflake into Star Schema

Initial dimension model - a snowflake



Physical Design Fundamentals

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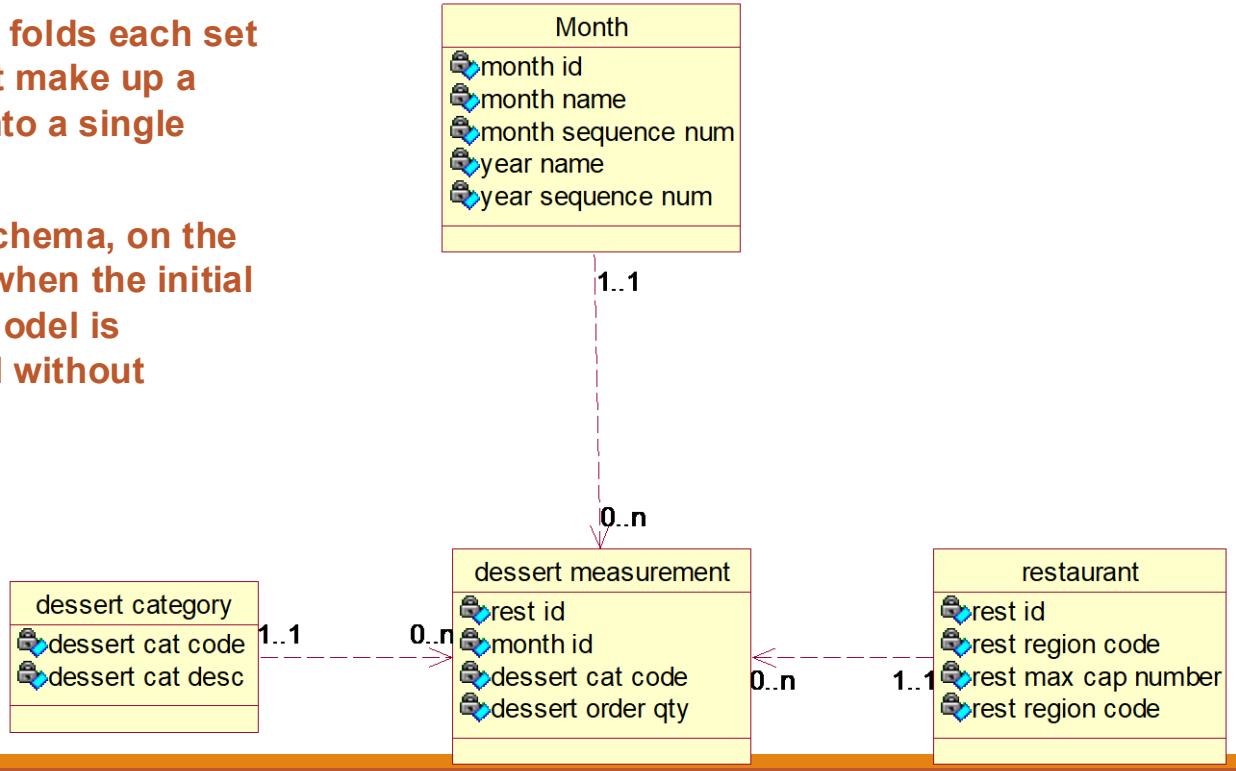
8. Use of Star Schema

Star Schema folds each set of tables that make up a dimension into a single table.

Snowflake schema, on the other hand, when the initial dimension model is implemented without "folding."